LULC Classification Prediction Using Remote Sensing Indices & Rapid Urbanisation Detection for Ludhiana, Punjab

Submitted by:

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In partial fulfilment of the requirements for the award of Master of Science in Geoinformatics



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22 December 2025

BONAFIDE CERTIFICATE

This is to certify that the project report entitled **"LULC Classification Prediction Using Remote Sensing Indices & Rapid Urbanisation Detection for Ludhiana, Punjab"** submitted by **Haripriya K R** **(Reg. No: 243308)** in partial fulfilment of the requirements for the award of **Master of Science in Data Analytics with Specialization in Geoinformatics** is a Bonafide record of the work carried out under our supervision.

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DECLARATION

I, Haripriya K R, a student of MSc Data Analytics with specialization in Geoinformatics, hereby declare that this report is substantially the result of my own work, except where explicitly indicated, and has been carried out during the period August 2025 – December 2025.

Place: DUK Haripriya K R

Date: December

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I am also deeply thankful to the distinguished faculty members and dedicated staff of the School of Digital Sciences, Digital University Kerala, for providing an excellent academic environment, indispensable resources, and constant encouragement, all of which have significantly contributed to the successful completion of this research endeavour.

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ABSTRACT

Rapid urbanisation is a major driver of land use and land cover (LULC) change in fast-growing Indian cities, leading to significant loss of vegetation and alteration of land surfaces. This study presents a LULC classification and rapid urbanisation analysis for Ludhiana, Punjab, using multi-temporal Sentinel-2 Surface Reflectance data processed in Google Earth Engine (GEE).

Dynamic World per-pixel land cover data were reclassified into four unified classes: Water, Vegetation, Built-up, and Barren. Multiple remote sensing indices like NDVI, EVI, SAVI, NDWI, MNDWI, NDBI, Urban Intensity (UI), and Bare Soil Index (BSI) were derived to enhance class discrimination. Classification accuracy was evaluated using confusion matrices, overall accuracy, and Kappa statistics.

Rapid urbanisation was assessed through density-based built-up mapping and pixel-wise LULC transition analysis between 2019 and 2024. Results indicate high classification accuracy (overall accuracy >0.87) and reveal that vegetation-to-built-up conversion (47.63 km^2) is the dominant land transformation, occurring mainly along the peri-urban fringe, while inner-city changes reflect redevelopment processes. The study demonstrates the effectiveness of index-based remote sensing for urban growth monitoring

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1.Introduction

Urbanisation is one of the most significant drivers of land use and land cover (LULC) change in developing countries, particularly in rapidly growing Indian cities. The expansion of urban areas alters natural landscapes, leading to the conversion of vegetated and agricultural land into impervious built-up surfaces. These changes have far-reaching environmental, socio-economic, and infrastructural implications, including loss of green cover, increased surface runoff, urban heat island effects, and pressure on water resources.

Monitoring and understanding LULC dynamics is therefore essential for sustainable urban planning and land management. Conventional ground-based surveys are often time-consuming, costly, and unable to capture rapid spatial changes. In contrast, **remote sensing and Geographic Information Systems (GIS)** provide a synoptic, consistent, and repeatable means of observing land surface changes over time. Recent advances in high-resolution satellite imagery, cloud-based processing platforms, and thematic land cover products have significantly enhanced the capability to perform large-scale and multi-temporal urban studies.

This study focuses on **Ludhiana**, one of the fastest-growing industrial cities in northern India, to analyse LULC patterns and detect rapid urbanisation using multi-temporal satellite data and remote sensing indices

1.1 Project Background and Significance

Ludhiana has undergone rapid urban growth over the past decade due to industrial development, population expansion, and continuous infrastructure enhancement. The city is characterized by a compact and nearly circular urban core, which is surrounded by predominantly rural and vegetated landscapes. Owing to limited availability of developable land within the city interior, recent urban growth has predominantly shifted towards the peri-urban fringe, where agricultural and vegetated areas are increasingly being converted into built-up land.

This outward expansion poses serious challenges, including the loss of fertile agricultural land, increased environmental stress, and unplanned spatial development at the city

periphery. Simultaneously, inner-city areas experience redevelopment and land restructuring, resulting in temporary and transitional land cover states that complicate accurate urban monitoring. Understanding both peripheral expansion and internal redevelopment is therefore crucial for capturing the complete dynamics of urban growth.

The integration of Sentinel-2 multispectral imagery, Dynamic World per-pixel land cover data, and remote sensing indices such as NDVI, NDBI, and Urban Intensity (UI) provides a robust framework for analyzing these complex land cover transitions. The use of Google Earth Engine enables efficient processing of large satellite datasets and supports consistent, scalable, and reproducible analysis. The significance of this study lies in its ability to accurately quantify recent LULC changes in Ludhiana, identify dominant land transformation processes, and provide spatially explicit insights that can support sustainable urban planning and land-use decision-making

1.2 Objectives

1. To classify land use and land cover (LULC) of Ludhiana, Punjab using multi-temporal Sentinel-2 satellite data.
2. To utilize remote sensing indices for improving discrimination between water, vegetation, built-up, and barren land cover classes.
3. To analyze spatial and temporal changes in LULC between 2019 and 2024.
4. To detect and quantify rapid urbanisation patterns in Ludhiana.
5. To identify dominant land cover transitions contributing to urban expansion, particularly at the peri-urban fringe.
6. To assess the accuracy and reliability of the LULC classification results.

1.3 Study Area



Fig 1.1 The figure shows the geographical location and administrative boundary of Ludhiana city, delineated by the highlighted outline

The study area is Ludhiana city, located in the central part of Punjab, northern India, at approximately 30.85° N latitude and 75.85° E longitude. Ludhiana is one of the major urban and industrial centres in the state and plays a significant role in regional manufacturing, trade, and transportation. The city is well connected through a network of national and state highways, contributing to its urban growth potential.

Spatially, Ludhiana is characterized by a compact and nearly circular urban core, surrounded by predominantly rural and agricultural land. This clear urban–rural interface provides a suitable setting for examining land use and land cover dynamics using remote sensing techniques. The surrounding landscape consists of extensive agricultural fields interspersed with settlements, canals, and transportation infrastructure, creating a heterogeneous land cover environment.

The terrain of the study area is largely flat, with minimal topographic variation, which reduces terrain-induced spectral distortion in satellite imagery and supports reliable multispectral analysis. The presence of built-up surfaces, vegetation, water bodies, and exposed soil provides distinct spectral responses, enabling effective LULC classification using Sentinel-2 imagery and derived indices.

Climatically, Ludhiana experiences a subtropical climate, with seasonal variations that influence vegetation cover and surface reflectance. These temporal variations make the region suitable for multi-temporal satellite-based analysis of land cover change. Overall, the spatial structure and land cover composition of Ludhiana make it an appropriate study area for investigating LULC classification and urban growth dynamics using geospatial methods.

2. Literature Review

Land use and land cover (LULC) change has been extensively studied as a key indicator of urban growth, environmental transformation, and landscape dynamics. Remote sensing and Geographic Information Systems (GIS) have become the primary tools for LULC analysis due to their ability to provide consistent, multi-temporal, and spatially comprehensive observations over large areas. Numerous studies have demonstrated the effectiveness of satellite-based monitoring for detecting urban expansion and associated land cover transitions.

Multispectral satellite imagery from sensors such as Landsat and Sentinel-2 has been widely used for LULC classification. Sentinel-2, in particular, offers improved spatial resolution and enhanced spectral capabilities, including red-edge and shortwave infrared bands, which support more accurate discrimination between vegetation, built-up areas, water bodies, and bare soil. Surface reflectance products derived from Sentinel-2 have further improved classification reliability by reducing atmospheric and radiometric distortions.

Remote sensing indices have long been employed to enhance thematic mapping by emphasizing specific land surface characteristics. Vegetation indices such as the Normalized Difference Vegetation Index (NDVI), Enhanced Vegetation Index (EVI), and Soil Adjusted Vegetation Index (SAVI) are commonly used to assess vegetation presence and condition. Water indices such as the Normalized Difference Water Index (NDWI) and Modified NDWI (MNDWI) have been shown to effectively delineate surface water features, particularly in urban environments. Similarly, built-up and soil-related indices, including the Normalized Difference Built-up Index (NDBI),

Urban Intensity indices, and Bare Soil Index (BSI), have been applied to highlight impervious surfaces and exposed soil.

In much of the existing literature, these indices are primarily used for threshold-based extraction, visual interpretation, or post-classification analysis, while LULC classification itself is often performed using raw spectral bands. However, recent studies have begun to explore the use of index-based feature sets as inputs to supervised machine learning classifiers, suggesting that indices can improve class separability and reduce spectral confusion in heterogeneous urban landscapes.

Machine learning techniques, particularly ensemble classifiers such as Random Forest, have gained prominence in LULC studies due to their robustness, ability to model non-linear relationships, and resistance to overfitting. Random Forest classifiers have been reported to outperform traditional parametric methods, especially in complex urban environments where spectral overlap between land cover classes is common.

The emergence of cloud-based geospatial platforms such as Google Earth Engine has further transformed LULC analysis by enabling efficient processing of large satellite archives and facilitating reproducible workflows. In parallel, global thematic datasets such as Dynamic World provide near-real-time, per-pixel land cover information derived from deep learning models and are increasingly being used as reference data for training and validation purposes.

Despite these advancements, several gaps remain in the literature. Index-driven machine learning–based LULC classification approaches are still comparatively limited, particularly in studies that focus on urban transition dynamics rather than static land cover mapping. Additionally, medium-sized Indian cities are underrepresented in urban remote sensing research when compared to large metropolitan regions. There is a need for integrated studies that combine remote sensing indices, supervised machine learning, cloud-based processing, and transition analysis to better understand recent urban growth patterns.

In this context, the present study contributes to existing research by employing a Random Forest classifier trained on index-based features derived from Sentinel-2 imagery, using Dynamic World data as labeled reference information, to analyze LULC dynamics and rapid urbanisation in Ludhiana, Punjab.

3. Materials and Methods

3.1 Data Acquisition

This study employs multi-temporal satellite and thematic datasets to analyze land use and land cover (LULC) dynamics and rapid urbanisation in Ludhiana, Punjab. The primary dataset used is Sentinel-2 Surface Reflectance (SR) imagery, acquired for the years 2019 and 2024. Sentinel-2 provides multispectral data at a spatial resolution of 10 m, suitable for detailed urban-scale LULC mapping.

To support supervised classification, Dynamic World land cover data were used as labeled reference information. Dynamic World provides per-pixel land cover labels derived from Sentinel-2 imagery using a deep learning–based classification framework. Dynamic World data corresponding to the period 2021–2023 were extracted and reclassified into four unified classes: Water, Vegetation, Built-up, and Barren.

Administrative boundary data for Ludhiana city were obtained from publicly available geospatial sources and used to delineate the study area. All datasets were processed within the Google Earth Engine (GEE) cloud computing platform.

3.2 Pre-processing of Satellite Data

Sentinel-2 Surface Reflectance imagery was pre-processed within Google Earth Engine prior to analysis. Cloud-affected pixels were removed using quality assessment bands, and images were spatially clipped to the Ludhiana city boundary. Multi-temporal composites were generated to reduce noise and seasonal variability. All input datasets were harmonized to a common spatial resolution and projection to ensure consistency across analyses.

3.3 Remote Sensing Indices Computation

To enhance discrimination between land cover classes, multiple remote sensing indices were derived from Sentinel-2 imagery. Vegetation characteristics were represented using NDVI, EVI, and SAVI, water features were highlighted using NDWI and MNDWI, built-up areas were emphasized using NDBI and Urban Intensity (UI), and exposed soil was characterized using the Bare Soil Index (BSI). These indices were selected based on their established effectiveness in urban and land cover studies and were used as feature inputs for LULC classification.

3.4 LULC Classification Using Random Forest

LULC classification was performed using a Random Forest (RF) classifier, implemented within Google Earth Engine. The Random Forest model was trained using the derived remote sensing indices as input features, while Dynamic World land cover data served as labeled reference samples. The classification was carried out into four unified LULC classes: Water, Vegetation, Built-up, and Barren. Random Forest was selected due to its robustness, ability to handle non-linear relationships, and strong performance in heterogeneous urban environments.

3.5 Accuracy Assessment

The accuracy of the classified LULC maps was evaluated using confusion matrices, overall accuracy, and Kappa coefficient. Validation samples were compared against reference labels to assess class-wise agreement and overall classification reliability. Accuracy assessment was conducted separately for each study year to ensure temporal consistency of results.

3.6 Rapid Urbanisation Detection and LULC Transition Analysis

Rapid urbanisation was analyzed using two complementary approaches. First, density-based built-up mapping was performed to identify spatial patterns of urban concentration and expansion intensity. Second, pixel-wise LULC transition analysis was conducted between 2019 and 2024 to quantify dominant land cover conversions. This transition analysis enabled identification of major

land transformation processes contributing to urban growth, while minor transitions were excluded based on negligible spatial extent.

4.Results and Evaluation

4.1 LULC Classification Results

Land use and land cover (LULC) classification maps for 2019 and 2024 were generated for Ludhiana using Sentinel-2 Surface Reflectance imagery and an index-based Random Forest classification framework. The classified outputs represent four unified land cover classes like Water, Vegetation, Built-up, and Barren ,each visualized using distinct colour codes to facilitate spatial interpretation of land cover distribution and change.

In the classified maps, built-up areas are represented in red, corresponding to urban and industrial surfaces such as residential zones, commercial areas, road networks, and other impervious features. Vegetation is shown in green, indicating agricultural land, croplands, and other vegetated surfaces surrounding the city. Water bodies are depicted in blue, representing rivers, canals, and surface water features. Barren land is displayed in light brown or beige, reflecting exposed soil, fallow agricultural land, construction sites, and other transitional land surfaces.

The 2019 LULC map shows a compact and continuous built-up core at the centre of Ludhiana, surrounded by extensive vegetated areas. Built-up land is largely confined to the established urban region, while vegetation dominates the surrounding rural and agricultural landscape. Water bodies appear as continuous linear features, primarily associated with river channels and canal networks. Barren land is present as scattered patches, particularly near the urban fringe and in limited construction zones.

In the 2024 LULC map, a noticeable increase in built-up areas is observed when compared to 2019. The expansion of red pixels is most prominent along the outer margins of the urban core, indicating outward growth into surrounding vegetated land. The overall circular morphology of the city is maintained, suggesting peripheral expansion rather than irregular or leap-frog sprawl. Within the urban core, built-up areas remain dense and continuous, with minimal availability of open land for further internal expansion.

Urban growth patterns show a strong association with transportation infrastructure. Built-up expansion extends linearly along major roads and highway corridors, forming elongated

development patterns radiating outward from the city centre. Increased built-up density is observed near road junctions and transport corridors, highlighting the role of accessibility in guiding urban development. Scattered built-up patches along these corridors in peripheral areas indicate early-stage development that may consolidate into continuous urban zones over time.

Within the inner city, small and fragmented patches of barren land appear embedded within built-up areas, particularly in the 2024 map. These patches represent temporary construction sites, demolition zones, and redevelopment areas, rather than permanent land cover change. Such features indicate ongoing urban restructuring and infrastructure development within the existing city fabric, rather than inward spatial expansion.

Vegetation cover surrounding Ludhiana shows increasing fragmentation from 2019 to 2024. Large, continuous vegetated patches observed in 2019 are increasingly subdivided in 2024 due to the expansion of built-up land at the urban periphery. This fragmentation is most evident along the urban–rural interface, where land cover transitions are actively occurring.

The river and surface water network, represented by blue pixels, remains spatially continuous in both classification years, with no evidence of major channel realignment or permanent loss of water bodies. However, a change in river width (thickness of blue pixels) is observed between 2019 and 2024, particularly along river margins. These variations are not spatially uniform and are interpreted as being primarily due to seasonal water level fluctuations, differences in image acquisition timing, and temporary exposure of riverbanks or sediments, rather than long-term hydrological change or direct urban encroachment. The presence of intermittent barren land pixels along river edges further supports the interpretation of seasonal or short-term surface water variability.

Overall, the LULC classification maps effectively capture the spatial structure of Ludhiana and its recent land cover changes. The results clearly depict a transition from a compact urban core toward outward expansion into surrounding vegetated areas, guided largely by transportation infrastructure, while inner-city areas exhibit localized redevelopment. The clear spatial separation between built-up, vegetation, water, and barren classes demonstrates the effectiveness of the index-based classification approach in representing urban land cover dynamics.

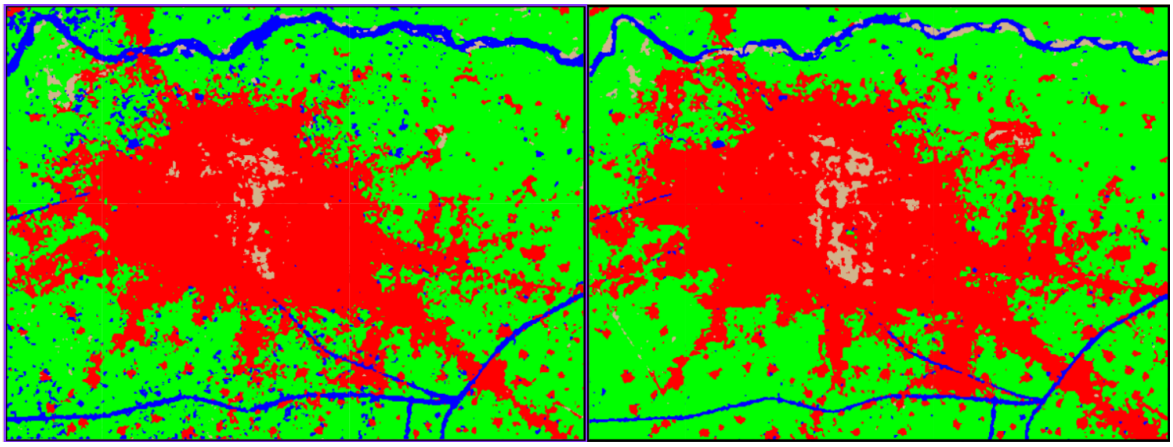


Fig 3.1

Land Use and Land Cover (LULC) classification maps of Ludhiana for the years 2019 and 2024.

4.2 Accuracy Assessment

The accuracy of the Land Use and Land Cover (LULC) classification results was evaluated to assess the reliability and consistency of the classified maps for 2019 and 2024. Accuracy assessment was carried out using confusion matrices, overall accuracy, and the Kappa coefficient, which are widely accepted statistical measures for evaluating supervised classification performance in remote sensing studies.

For the 2019 classification, the overall accuracy was 0.88, with a corresponding Kappa coefficient of 0.84. These values indicate a strong agreement between the classified LULC map and the reference data, suggesting that the Random Forest classifier effectively discriminated between the four land cover classes. The high Kappa value confirms that the observed agreement is substantially higher than agreement expected by chance.

Similarly, the 2024 classification achieved an overall accuracy of 0.87 and a Kappa coefficient of 0.832, demonstrating consistent classification performance across time. Although a slight reduction in accuracy is observed when compared to 2019, the values remain well within acceptable limits for urban land cover studies and indicate reliable model generalization under increased landscape complexity.

Analysis of the confusion matrices reveals that the Water and Vegetation classes exhibit high classification accuracy with minimal misclassification. These classes show strong spectral separability due to the use of water and vegetation indices such as NDWI, MNDWI, NDVI, EVI, and SAVI. In contrast, most classification errors occur between the Built-up and Barren classes. This confusion is expected in urban environments, as construction sites, exposed soil, and newly developed areas often exhibit spectral characteristics similar to impervious surfaces, particularly in the shortwave infrared bands.

The persistence of similar confusion patterns across both years suggests stable classifier behavior rather than random error. Importantly, the misclassification between Built-up and Barren land does not significantly affect the interpretation of overall urban growth trends, as both classes are associated with anthropogenic land transformation processes.

Overall, the accuracy assessment confirms that the index-based Random Forest classification approach produces robust and reliable LULC maps, suitable for subsequent analysis of urban expansion, density patterns, and land use transitions in Ludhiana.

4.3 Confusion Matrix Evaluation

The confusion matrix analysis provides a detailed evaluation of class-wise classification performance for both 2019 and 2024. The results indicate that the Water and Vegetation classes achieve high classification accuracy with minimal misclassification. These classes exhibit strong spectral separability due to distinct reflectance characteristics, which are effectively captured using water and vegetation indices such as NDWI, MNDWI, NDVI, EVI, and SAVI.

The majority of classification errors are observed between the Built-up and Barren classes. This confusion arises primarily from their spectral similarity in areas dominated by exposed soil, construction activity, and transitional land surfaces. Such confusion is commonly reported in urban LULC studies and is characteristic of rapidly developing cities where land is frequently under conversion. Importantly, this misclassification does not significantly affect the interpretation of overall urban growth trends, as both classes represent anthropogenic land transformation processes.

The similarity in confusion patterns across both study years indicates stable model performance and consistent class discrimination. This temporal consistency confirms that the classifier

generalizes well across time and that observed land cover changes reflect actual surface dynamics rather than classification instability

Actual \ Predicted	Water (1)	Vegetation (2)	Bare (3)	Built-up (4)
Water (1)	**1930**	38	23	53
Vegetation (2)	32	**1717**	134	48
Bare (3)	15	159	**1631**	151
Built-up (4)	65	26	190	**1686**

Actual \ Predicted	Water (1)	Vegetation (2)	Bare (3)	Built-up (4)
Water (1)	**1902**	12	27	34
Vegetation (2)	13	**1865**	122	57
Bare (3)	27	198	**1610**	213
Built-up (4)	42	54	213	**1691**

Fig 4.1 Confusion matrix of 2019 and 2024 LULC classification

4.4 Rapid Urbanisation Detection

Rapid urbanisation in Ludhiana was analysed by explicitly identifying and examining pixels that transitioned to built-up land between 2019 and 2024. Pixels classified as non–built-up in 2019 and built-up in 2024 were isolated to represent newly developed urban areas. This change-based approach ensures that the analysis focuses solely on recent urban expansion, rather than the overall extent of existing built-up land.

The extracted built-up change pixels were subjected to density-based spatial analysis to evaluate the concentration and intensity of recent urban growth. The resulting urbanisation density map classifies newly urbanised areas into multiple density categories, ranging from low-density scattered development to very high-density clustered expansion, based on the spatial concentration of change pixels.

Low-density and medium-low-density classes represent isolated or weakly clustered new built-up pixels, indicating dispersed or early-stage development, primarily located toward the outer peri-urban regions. Medium-density zones indicate moderate clustering of new development and correspond to transitional areas where urban expansion is actively progressing but not yet continuous.

Medium-high, high, and very high-density classes represent strong clustering of newly built-up pixels, indicating zones of intensive and coordinated urban expansion. These high-density clusters are predominantly concentrated along the outer margins of the existing urban core, forming distinct expansion fronts. The spatial distribution of these clusters shows a strong alignment with major transportation corridors, including highways and arterial roads, highlighting the role of accessibility and infrastructure in guiding recent urban growth.

Within the inner city, the presence of built-up change pixels is minimal, indicating limited scope for further spatial expansion inside the urban core. Changes within the core are largely associated with redevelopment and restructuring processes, which do not result in net addition of built-up land and are therefore not prominent in the change-pixel density analysis.

Overall, the change-pixel-based density analysis provides a direct and process-oriented representation of rapid urbanisation, clearly distinguishing between concentrated expansion zones and dispersed peripheral development. This approach enables precise identification of active urban growth fronts and offers a robust framework for assessing recent urbanisation dynamics in Ludhiana.

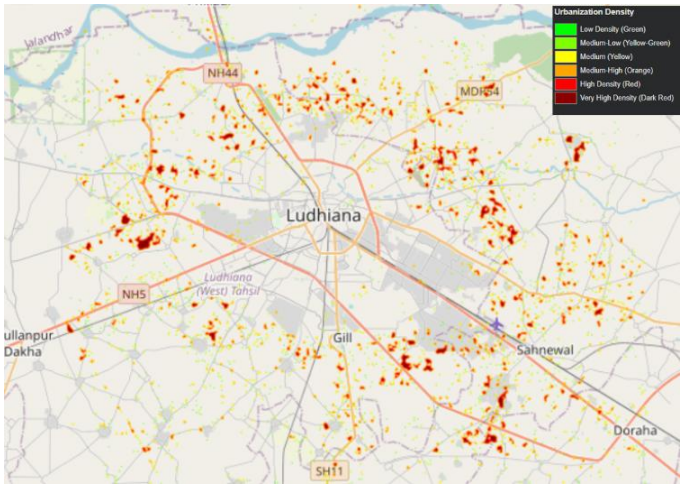


Fig 4.2 Spatial density of newly urbanised pixels in Ludhiana (2019–2024)

4.5 LULC Transition Analysis (2019–2024)

Pixel-wise Land Use and Land Cover (LULC) transition analysis was conducted to quantify land cover changes between **2019 and 2024**. Each pixel was tracked across the two time periods to identify its transition from one LULC class to another. Among all detected transitions, **five dominant transition types** were identified based on spatial extent, while the remaining transitions were found to be negligible and were therefore excluded from detailed analysis. The identified transitions represent the major land transformation processes associated with recent urbanisation in Ludhiana.

Vegetation → Built-up (47.6309 km²)

This transition represents the conversion of vegetated and agricultural land into built-up areas and is the most dominant land cover change observed during the study period. Spatially, this transition is concentrated along the peri-urban fringe of Ludhiana and occurs in relatively contiguous patches. The pattern indicates systematic outward urban expansion into surrounding rural and agricultural land. The large spatial extent of this transition confirms that vegetation loss is the primary driver of recent urban growth in the city and reflects permanent land transformation associated with residential, industrial, and infrastructural development.

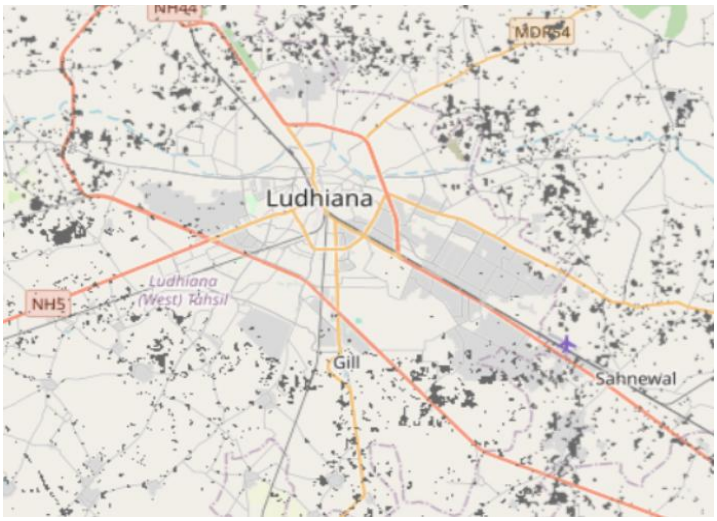


Fig 4.3 Vegetation to Built-up visualisation (grey)

Built-up → Vegetation (6.4554 km²)

The built-up to vegetation transition occurs over a much smaller area and is spatially scattered across the study region. This transition does not form large contiguous clusters and is not dominant in any particular zone. It likely represents localized and temporary land cover changes such as vegetation growth on vacant urban plots, development of small green spaces, or seasonal vegetation effects influencing spectral classification. This transition does not indicate large-scale urban contraction but reflects localized variability within the urban and peri-urban environment.

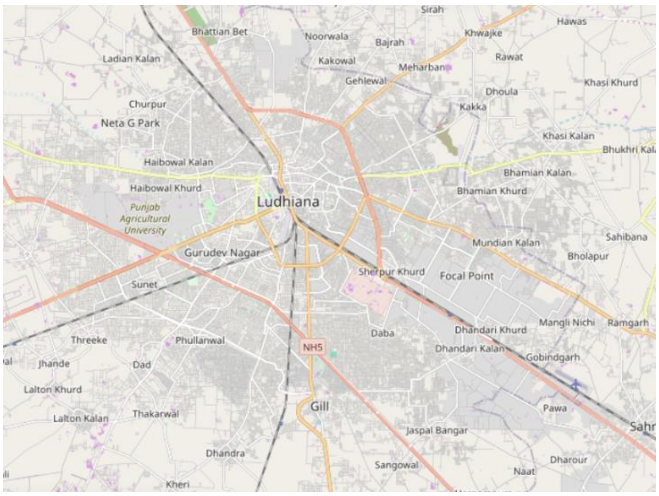


Fig 4.4 Built-up to Vegetation visualisation (pink)

Water → Barren (5.1396 km²)

The water to barren transition is primarily observed along riverbanks and surface water margins, appearing as narrow, linear spatial features. This transition is not associated with direct urban development. Instead, it is interpreted as the result of seasonal water level fluctuations, exposure of sediments, or variation in river width due to differences in image acquisition periods. The absence of widespread conversion from water to built-up land suggests that this transition represents temporary or seasonal hydrological variation rather than permanent water loss.



Fig 4.5 Water to Barren Visualisation (blue)

Vegetation → Barren (4.3032 km²)

The vegetation to barren transition represents areas where vegetative cover has been removed, leaving exposed soil or fallow land. This transition is mainly distributed in peri-urban and transitional zones, rather than within the dense urban core. Such changes are commonly associated with land clearing, fallow agricultural cycles, or pre-construction activities. In several cases, this transition may act as an intermediate stage preceding further conversion to built-up land

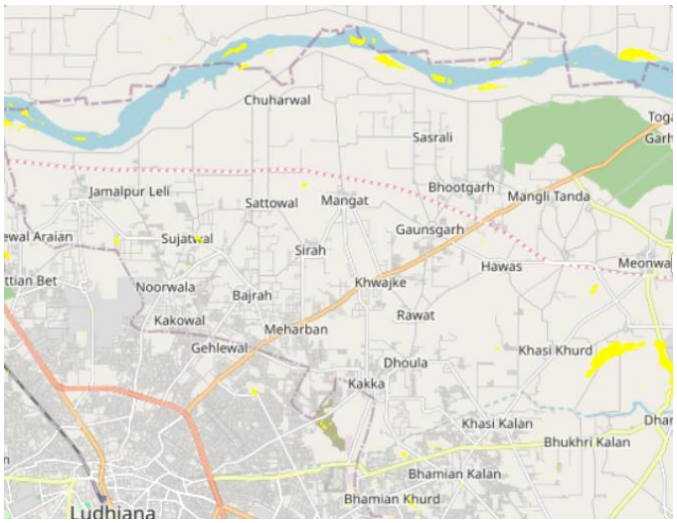


Fig 4.6 Vegetation to Barren visualisation (yellow)

Built-up → Barren (4.1678 km²)

The built-up to barren transition is predominantly observed within the existing urban area and appears as small, fragmented patches. This transition reflects urban redevelopment processes such as demolition of old structures, construction staging, and infrastructure upgrades. It does not represent reversal of urbanisation but highlights the dynamic and transitional nature of land cover within established urban environments.



Fig 4.6 Built-up to Barren visualisation (grey)

5.Conclusion

This study examined land use and land cover (LULC) dynamics and rapid urbanisation in Ludhiana, Punjab, using multi-temporal Sentinel-2 Surface Reflectance data processed in Google Earth Engine. An index-based Random Forest classification approach, trained using Dynamic World per-pixel land cover labels, was employed to generate LULC maps for 2019 and 2024. The high overall accuracy and Kappa agreement obtained for both years confirm the reliability and robustness of the adopted classification framework for urban-scale LULC analysis.

The results clearly indicate that recent land cover change in Ludhiana is dominated by vegetation-to-built-up conversion, highlighting outward urban expansion as the primary mode of urban growth. Pixel-wise LULC transition analysis demonstrates that this conversion accounts for the largest share of land transformation, while other transitions such as built-up to barren, vegetation to barren, and water to barren represent localized, transitional, or seasonal processes rather than permanent land cover change. These findings confirm that urbanisation in Ludhiana is largely driven by conversion of surrounding vegetated and agricultural land rather than by inward densification of the urban core.

Rapid urbanisation analysis based on change-pixel density mapping provided additional insight into the spatial intensity of recent urban growth. Newly built-up pixels were found to be concentrated along the peri-urban fringe, forming distinct expansion fronts, while low-density clusters represent dispersed or early-stage development further toward the outskirts. The alignment of high-density expansion zones with major transportation corridors indicates that infrastructure accessibility plays a key role in guiding the spatial direction of urban growth. In contrast, the urban core shows minimal presence of newly developed built-up land, suggesting limited scope for further spatial expansion within the existing city fabric and indicating that changes within the core are largely associated with redevelopment processes.

The LULC classification maps also reveal that water bodies remain largely spatially stable over the study period. Observed variations in river width are interpreted as seasonal or short-term surface water fluctuations influenced by image acquisition timing, rather than permanent hydrological alteration or direct urban encroachment. Minor misclassification between built-up and barren land was observed, particularly in construction and transitional zones, which is a known limitation in urban remote sensing studies due to spectral similarity between exposed soil and impervious surfaces.

Overall, the integrated use of remote sensing indices, machine learning classification, Dynamic World reference data, and cloud-based geospatial processing provides an effective and scalable framework for monitoring urban growth and LULC dynamics. While the analysis is limited to discrete temporal snapshots and may not capture short-term intra-annual variations, the methodology successfully captures both the magnitude and spatial pattern of recent urbanisation. Future work may benefit from higher temporal resolution analysis, integration of SAR data to reduce spectral ambiguity, and incorporation of socio-economic variables to support predictive urban growth modeling. The approach demonstrated in this study can be readily extended to other medium-sized Indian cities to support urban monitoring and sustainable land-use planning.

6.Reference

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